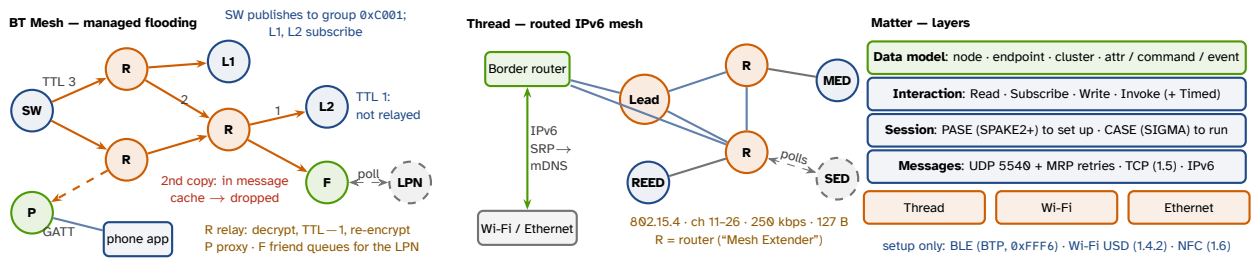


In one line: **Bluetooth Mesh** floods tiny double-encrypted messages over BLE *advertising* — no IP, no connections; **Thread** is a **routed IPv6** mesh on 802.15.4 that reaches the LAN through **border routers**; **Matter** is the **application layer** — data model, PASE / CASE, attestation, fabrics — over Thread, Wi-Fi or Ethernet, set up over BLE (or Wi-Fi USD, NFC).



Bluetooth Mesh — SIG Mesh Protocol 1.1

IVI	NID	CTL	TTL	SEQ	SRC	DST	Transport PDU	NetMIC
1	7	1	7	24	16	16	(AppKey-sealed)	32 / 64

Network PDU, bits. grey clear (NID picks the key) · orange obfuscated (privacy key) · green AES-CCM encrypted under a NetKey-derived key · MIC 64 bit for control PDUs

unicast 0x0001–0xFFFF one per element	virtual 0x8000– hash of Label UUID	group 0xC000– 0xFFFF	fixed 0xFFF00–
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fixed groups: FFFB directed-forwarding nodes · FFFC proxies · FFFD friends · FFEE relays · FFFF all

- Bearer:** non-connectable adverts — AD 0x2A message · 0x2B beacon · 0x29 PB-ADV; nodes scan all the time. GATT only through a **Proxy**: service 0x1828 (provisioning 0x1827, solicitation 0x1859). No links, no LE pairing.
- Managed flooding:** TTL 7 bits — 0 never relayed, arriving as 1 not relayed again, 2–127 relayable; a message cache drops repeats. **Directed forwarding** (1.1) discovers paths per subnet and mixes with flooding.
- Node** → **elements** (one unicast address each) → **models**: a client *publishes* to an address, servers *subscribe* — a group is a topic.
- Features:** Relay · Proxy · Friend (queues for its LPN) · **Low Power** (sleeps, polls its Friend; PollTimeout 1 s ... ≈ 96 h in 100 ms steps).
- Keys,** AES-128-CCM twice: **NetKey** (network layer, ≤ 4096 subnets — relays forward but cannot read) · **AppKey** (payload, bound per model, ≤ 4096) · **DevKey** (per node, from the provisioning ECDH, never on air).
- Size:** unsegmented access message ≤ 11 B; segmented 12 B × ≤ 32 = 384 B upper transport → ≤ 380 B (32-bit TransMIC) / 376 B. Control, not bulk.
- Replay:** SEQ 24 bits per element + 32-bit **IV Index**; a persisted replay list per source. SEQ lasts 2.66 years at one message / 5 s, 19 days at / 100 ms, then IV Update via beacons (up to 96 h to spread; started only by primary-subnet nodes).
- Provisioning** (PB-ADV / PB-GATT, often a phone): beacon with 128-bit Device UUID → algorithm choice → P-256 ECDH (device key in band, or OOB / X.509) → OOB auth → NetKey + unicast address; both sides derive the DevKey.

active MITM?	1.0, key in band	1.1, key in band	key OOB
no OOB auth	vulnerable	vulnerable	none known
static / output / input	vulnerable	none known if enough entropy	none known

Entropy: 5 alphanumeric ≈ 25 bits, 8 ≈ 41; static OOB up to 256 bits (1.0: 128; its flaws CVE-2020-26556/-57/-59/-60). **Key Refresh:** distribute (send old, accept both) → send new → revoke old; excluded nodes are cut off — do it before discarding a node ("trash-can attack"). **1.1 adds:** directed forwarding · remote provisioning (+ device-key refresh) · certificate-based provisioning · private beacons · subnet bridge · firmware update (BLOB transfer).

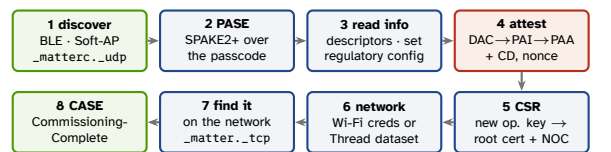
Thread

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
Router ID · 6 bits						R	Child ID · 9 bits (0 = the router)									RLOC16: route by top 6 bits	

- Radio:** IEEE 802.15.4 (2006 / 2015), 2.4 GHz O-QPSK, channels 11–26, 250 kbit/s, ≤ 127-byte frames, CSMA-CA, MAC acks + retries, MAC-layer encryption. **6LoWPAN** (RFC 4944 / 6282) compresses and fragments IPv6 (min MTU 1280); MLE replaces neighbour discovery.
- Roles:** Router ("Mesh Extender" in 1.4) · **Leader** (one per partition, hands out router IDs, re-elected) · REED (promotes below 16 routers) · FED · MED (radio on) · SED (polls its parent) · SSED (listens on a schedule) · Border Router (any FTD, several allowed). All attach as a child first (MLE Parent / Child ID Request).
- Routing:** proactive distance-vector (like RIPng) in MLE adverts; ≤ 32 active routers (64 IDs, kept at 16–23), ≤ 511 children each. Link quality 3 / 2 / 1 / 0 → cost 1 / 2 / 4 / ∞; path cost = minimum sum.
- Addresses:** mesh-local ULA / 64 — **ML-EID** (stable, for apps) + **RLOC** (moves with topology) + ALOC anycast; BR prefixes via SLAAC / DHCPv6. **Dataset:** channel, PAN ID (2B), XPAN ID (8B), name, prefix, Network Key, PSKc; joining = DTLS Joiner → Commissioner via a Joiner Router.
- Border router:** IPv6 both ways; devices register services by unicast **SRP**, the BR re-advertises them over **mDNS**; Border Agent for outside commissioners.
- 1.3.0** (19 Jul 2022): standard BRs, SRP / DNS-SD — what Matter needs. **1.4** (Sep 2024): **credential sharing** (BR shows a 9-digit one-time code → ePSKc → DTLS → dataset; required in 1.4 BRs) · NAT64 / DNS64 + DHCPv6-PD · diagnostics · TREL (802.15.4 frames in UDP/IPv6 over Wi-Fi / Ethernet) · TCAT (TLS + X.509 over BLE, bulk setup).

Matter — CSA application layer

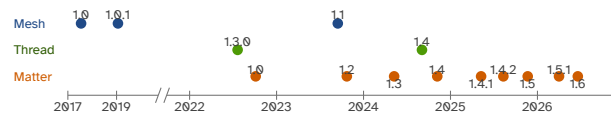
1.0: 4 Oct 2022. Zigbee, Z-Wave, BT Mesh: only via a **bridge** (bridged devices = endpoints).



MT:	ver	VID	PID	flow	caps	discrimin.	setup passcode	pad
	3	16	16	2	8	12	27	4

QR payload, bits, base-36 text. Manual code: **11 digits** = 4-bit short discriminator + passcode + Verhoeff check (21 with VID / PID). Passcode 00000001-99999998, never 11111111 ... 88888888, 12345678, 87654321.

- Data model:** **node** (64-bit node ID per fabric) → **endpoints** (0 = root, utility clusters) → **clusters** (server holds state, client does not) → attributes · commands (RPC) · events (past changes); paths may use groups and wildcards.
- Interaction:** Read · Subscribe (reports) · Write · Invoke; **Timed Write** / Invoke for locks and garage doors (anti-replay); groups = IPv6 multicast.
- Sessions:** PASE, commissioning only. CASE = SIGMA (Sigma1–3 or resumption), P-256 ECDH + ECDSA, chain NOC → ICAC → RCAC. MRP: 4 resends (SDK default).
- Discovery:** BLE service 0xFFFF6 (BTP), adverts every 20–60 ms for 30 s then 150–1500 ms · Soft-AP SSID MATTER-ddd-vvvv-pppp · DNS-SD _matterc._udp (Thread via SRP). Operational: _matterc._tcp, instance <compressed fabric ID>-<nnode ID>.
- Attestation:** per-device DAC → PAI → PAA; PAAs trusted via the CSA's **Distributed Compliance Ledger**; Certification Declaration signed by a CSA key (VID, PID, development / provisional / official); signature over a nonce + the session's challenge.
- Fabric** = root CA + 64-bit fabric ID. **Multi-admin:** one node in several fabrics sharing one data model; spec minimum 5 fabrics (SDK default 16).
- Releases:** 1.2 nine appliance types · 1.3 energy reporting, EV chargers · 1.4 HRPAP (router + Thread BR), enhanced multi-admin, long-idle devices + Check-In · 1.4.1 multi-device QR, NFC tag · 1.4.2 Wi-Fi-only setup (USD), revocation lists, VID check, BRs ≥ 150 devices on Thread 1.4 · 1.5 cameras (WebRTC), closures, soil sensors, tariffs, TCP · 1.5.1 camera multi-stream · 1.6 NFC commissioning, **Joint Fabric**, thermostat suggestions.



On the phone

- Apple:** Matter in Home needs a **home hub**; Thread accessories a Thread hub (HomePod mini, HomePod 2nd gen, Apple TV 4K 3rd gen Wi-Fi + Ethernet) or a third-party BR. MatterSupport (iOS 16.1+); MatterAddDeviceRequest; its extension commissions into the app's own ecosystem. ThreadNetwork (iOS 15+); THClient shares Thread credentials.
- Android:** Play services CommissioningClient — Google fabric and / or the app's own (CommissioningService); Thread Network SDK; preferredCredentials, addCredentials (Android 8.0+; reaching end devices needs 8.1+).

Traps

- ! "Matter = Thread" — app layer vs one transport; no BR, no way off the mesh.
- ! A NetKey holder cannot forge app messages, but can flood, or fill replay lists with SEQ 0xFFFFFFFF so nodes ignore everyone until the next IV Update.
- ! Mesh starter kit with a fixed NetKey — the owner can never rekey; ship a provisioning database and unique keys instead.
- ! Mesh guests get their own NetKey + AppKey; a Key Refresh that skips them revokes them.
- ! A BR per vendor, each forming its own network: share Thread credentials.